HW 4 Marginal Rate of Technical Subs

Problem 7.1

A) Assuming that output sells for $3 per unit.

X Fixed at 2 units

Units of Y used	Total product of y	Marginal product of y	Avg of y	Marginal Revenue Product of y
1	130	130	130	390 $
2	188	58	94	174 $
3	234	46	78	138 $
4	272	38	68	114 $
5	305	33	61	99 $

$$MP = \frac{\Delta TP}{\Delta Input}$$

$$\text{Avg product} = \frac{TP}{\text{Units of input}}$$

$$NRP = MP \times P$$

MP_y $y = 3$

$\rightarrow 234 - 188 = 46$

$AP_y = \frac{234}{3} = 78$

$MRP_y = 46 \times 3 = 138\ \$$

y Fixed at 3 units

Units of X used	Total product of X	Marginal product of X	Avg Product of X	NRP
1	162	162	162	486 $
2	234	72	117	216 $
3	282	48	94	144 $
4	324	42	81	126 $
5	360	36	72	180 $

Problem 7.1

B) How many units of y should the company employ?

X = 2units, fixed

Output price = $3

Cost per unit of y = 120 $ per day

$MRP_y \geq MC_y$

Unit of y	MP	MRP at $3 per unit	Cost of y	Decision
1	130	390 $	$ 120	Employ
2	58	174 $	$ 120	Employ
3	46	138 $	$ 120	Employ
4	38	114 $	$ 120	NO
5	33	99 $	$ 120	NO

The third units adds 138 $ in revenue against of $ 120 cost.

Optimal is $y^* = 3$ units.

Marginal revenue of product the third unit exceeds its cost by $ 18. The fourth unit adds only $ 114 against ~~a cost of $ 120~~. reduce profit by $6.

Problem 7.1

Part C) Should the company change its current input combination

Assuming current input combination
$X = 1$, $y = 3$

$$TC = (1 \times 120) + (3 \times 120) = 480\ \$$$

Output = 162 units per day.

Cost per unit of $(X, y) = 120\ \$$

Current Revenue

$$TR = Q \times P = 162 \times 3 = 486\ \$$$

No othe costs

$$\pi = 486\ \$ - 480\ \$ = 6\ \$$$

Output lost by reducing y from 3 to 2
from production table
$162 - 130 = 32$

Output gained by increasing X from 1 to 2 while $y = 2$
Output increase from 130 to 188

output gained $= 188 - 130 = 58$

$\Delta Q = 58 - 32 = 26$ units

The new combination $X = 2$, $y = 2$ produces 188 units instead of 162 units at the same total input cost of 480 $

Part D)

Constant returns to scale.

Problem 7.8 Optimal input level

Given $Q = 5{,}000 + 100A - 0.5A^2$

Part A

$$MP_{Advertising} = \frac{\partial Q}{\partial A} = 100 - A$$

At: $A = 0$, MP is 100 tickets

$A = 20$, MP is 80 tickets

$A = 100$, MP is zero

diminishing marginal returns. as more ads are purchased, each ads generate fewer additional ticket sales.

Part B

Max-profit

$MRP = MRC$

$MRP = MP_A \times \$2$

$= 2(100 - A) = 200 - 2A$

$MRC_A = 100\ \$$

$MRP > MRC \Rightarrow$ employ more of the resource

$MRP < MRC \Rightarrow$ reduce

$MRP = MRC$ (Max-profit)

$200 - 2A = 100$

$200 - 100 = 2A$

$A = 50$

Part C

$MRP_A = 50 \times 2\$ = 100\$$

$MP_A = 100 - 50 = 50$ tickets

Therefore marginal benefit equals the marginal cost.

$$SOC = \frac{dQ}{dA} = -1 < 0$$

Expected ticket sales $A = 50$

$Q = 5{,}000 + 100(50) - 0.5(50)^2$

$Q = 5{,}000 + 5{,}000 - 1250$

$Q = 8{,}750$ tickets per event

Problem 7.9 Net Marginal Revenue (NMRP) rule to determine whether law firm should change its combination of partners and associates.

Given

	Partners	Associates
1) Client billing rate per billable hour	$250	$125
2) Billable hours per 40 hour week	25	35
3) Non billable hours per week	15	5
4) Variable overhead as % of revenue	50%	60%
5) Annual compensation	$175,000	$70,000
6) Working weeks per year	50	50

A) Partners

NMRP = MRP - variable overhead

Annual billable hours = 25 x 50 = 1,250 hrs

Annual revenue = 1,250 x $250 = $312,500

Variable overhead = $312,500 x 0.50 = $156,250

$NMRP_P$ = 312,500 - 156,250

= 156,250 $

Annual compensation = $175,000

Associates

35 x 50 = 1,750 hours

1750 x 125 = $218,750

$218,750 x 0.6 = $131,250

$NMRP_A$ = 218,750 - 131,250

= 87,500 $

(+)

$70,000

Part B

The firm should consider substituting associates for partners. It can maintain adequate supervision. However, the exact optimal staffing level cannot be determined from these figures alone.